University of Alkafeel, College of Dentistry, Year 3, Course 1

Pathology Lab 7

Hemodynamic Disorders

By: Assistant Lecturer, Soroor M. Hadi

Learning Objectives

- ❑Know what Hemodynamic Disorders are
- ❑Know what is Edema and its types
- ❑Know what Hyperemia and Congestion are.
- ❑Know what Hemorrhage is and its types

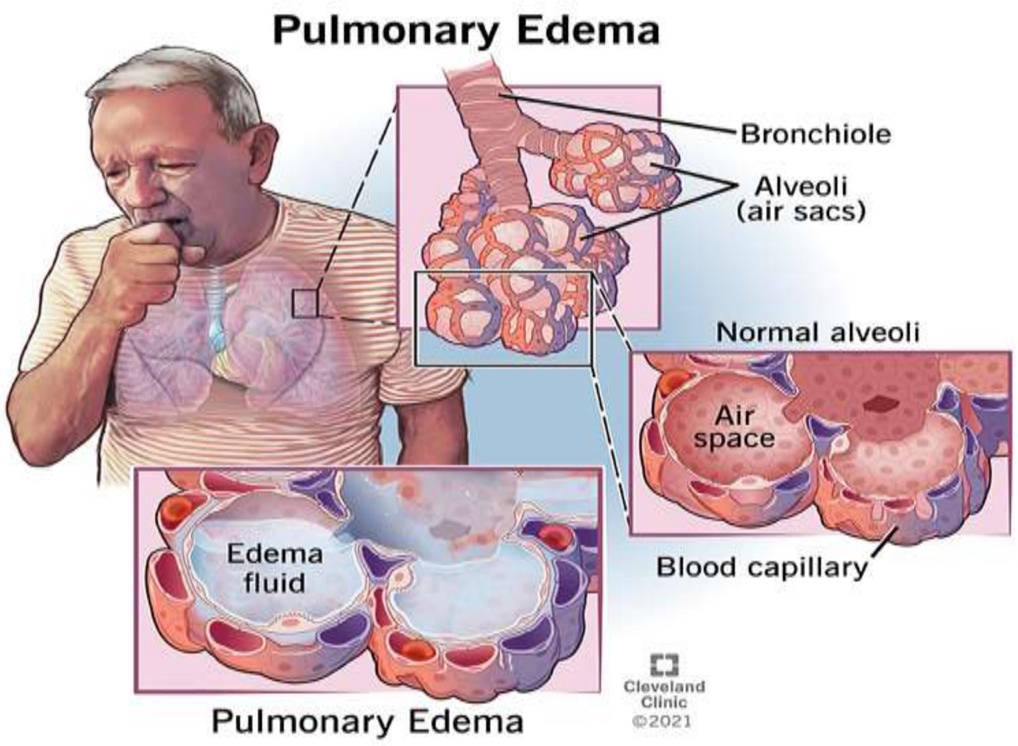
Introduction

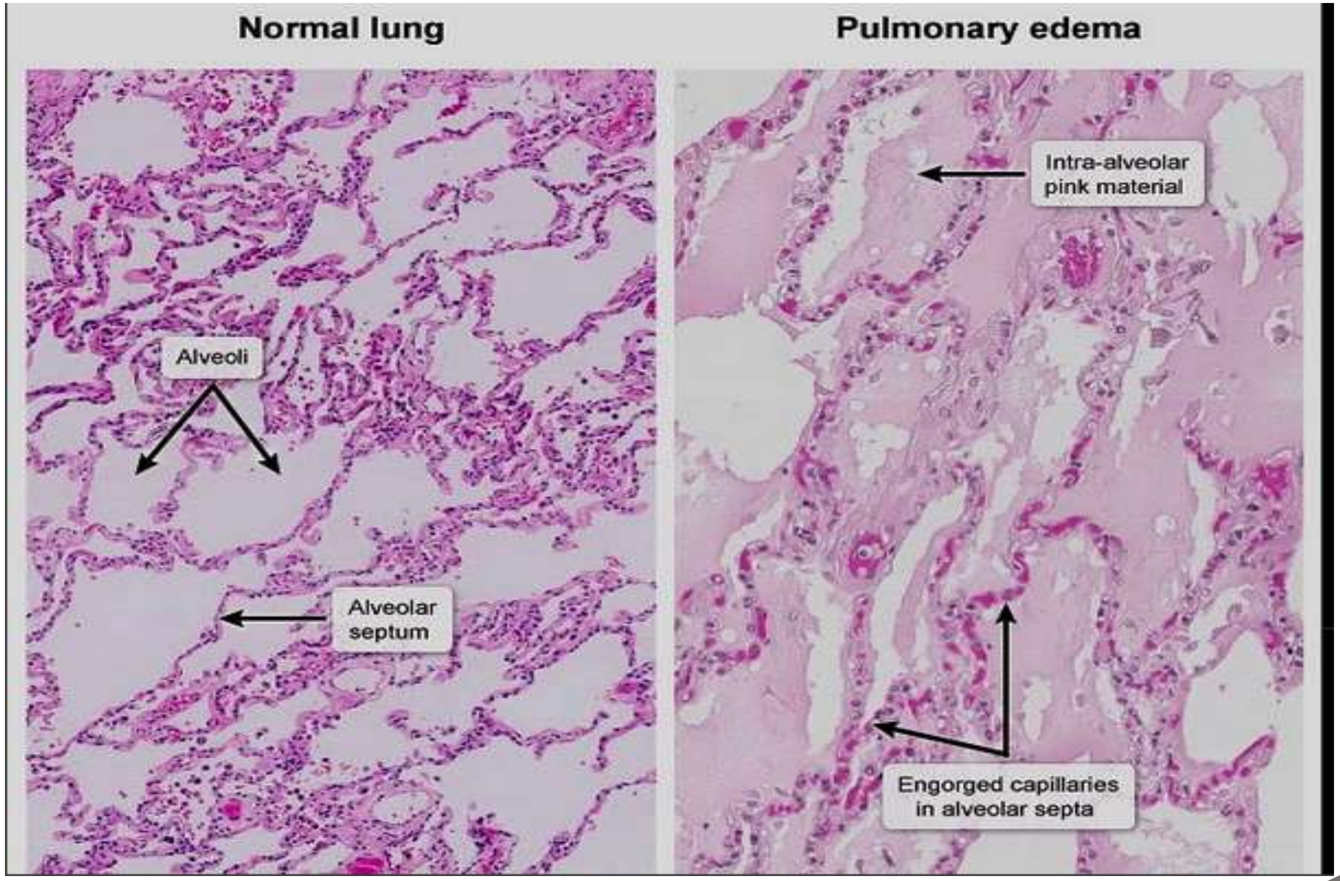
Hemodynamic disorders are a group of conditions that result from disturbances in the **volume**, **distribution**, or **flow of blood**.

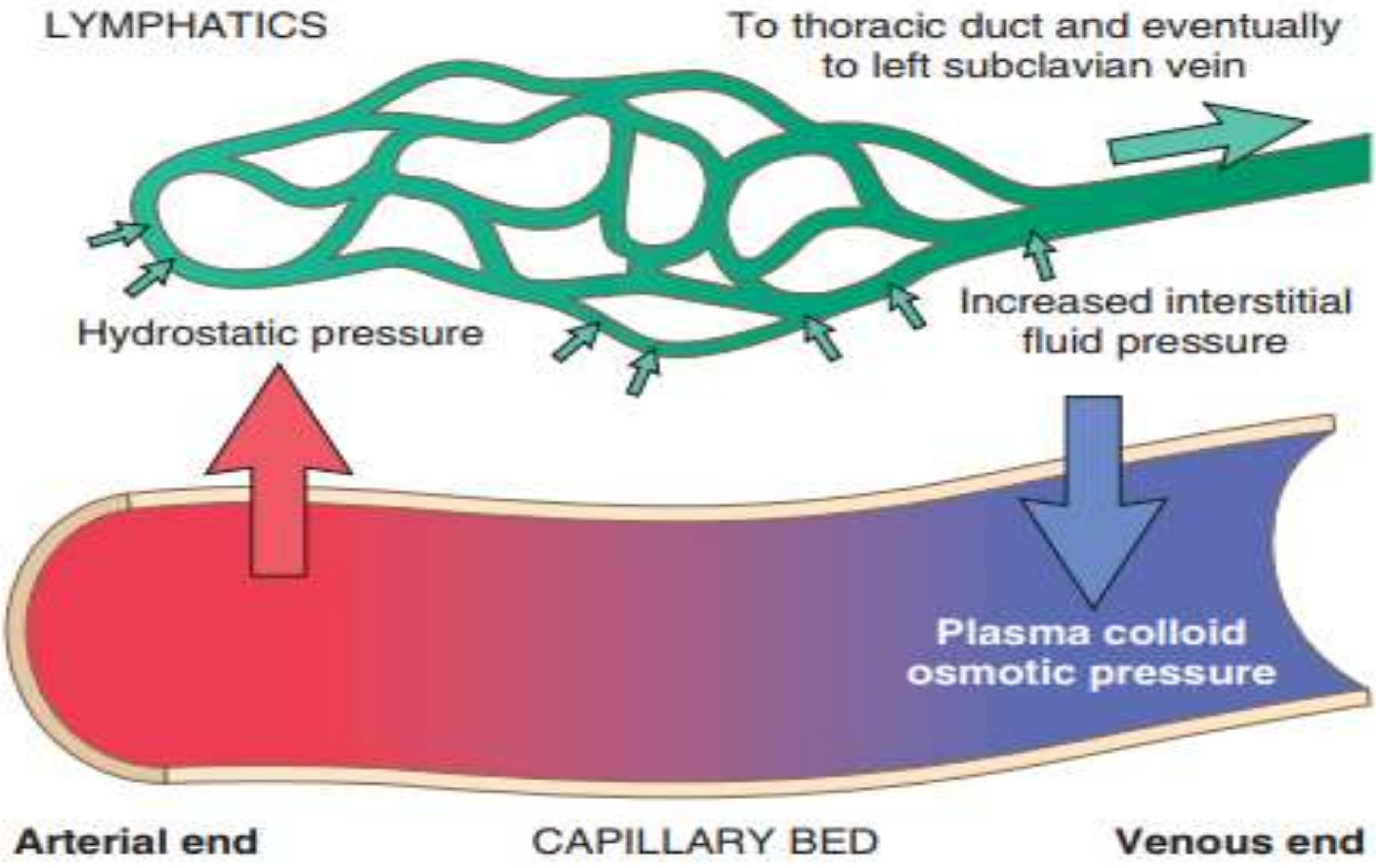
These disturbances disrupt the essential balance between fluid in the blood vessels and fluid in the tissues, leading to a cascade of pathological consequences, from **localized swelling** to **systemic organ failure.**

Edema

Edema is the accumulation of excessive fluid in the interstitial tissue spaces.






LYMPHATICS
To thoracic duct and eventually to left subclavian vein
Hydrostatic pressure
Increased interstitial fluid pressure
Plasma colloid osmotic pressure
Arterial end
CAPILLARY BED
Venous end

Categories of Edema

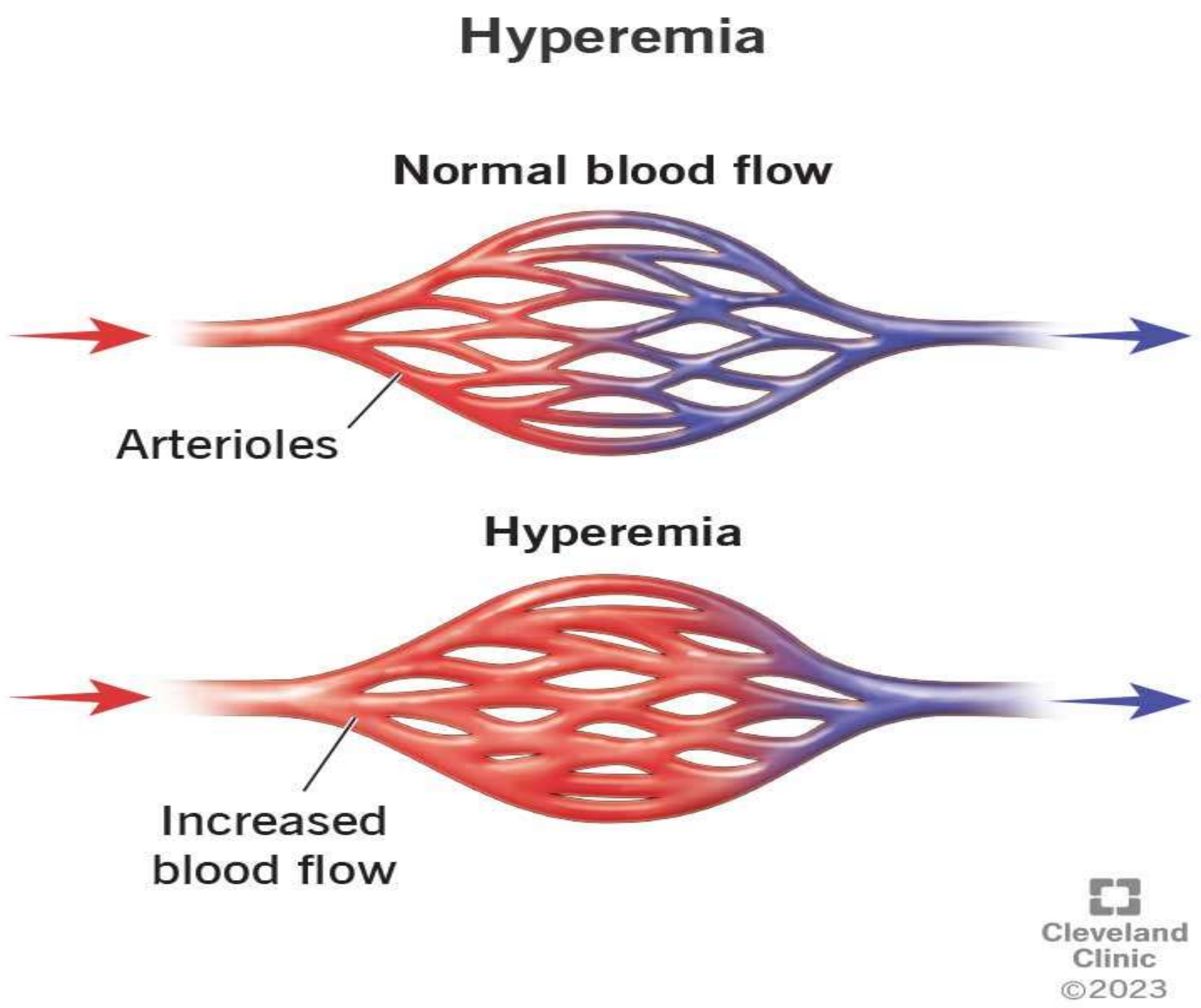
Category	Cause of Imbalance	Example Disorder	Edema Type
Increased Hydrostatic Pressure (↑Pushing pressure)	Impaired venous return or arteriolar dilation	Deep Vein Thrombosis , Heart Failure.	**Transudate** (protein-poor)
Decreased Oncotic Pressure (↓Pulling Pressure)	Reduced albumin synthesis or increased albumin loss.	Nephrotic Syndrome (proteinuria); Liver Cirrhosis (reduced synthesis); Malnutrition.	**Transudate**
Lymphatic Obstruction	Impaired lymphatic drainage (lymphatics normally clear small amounts of leaked protein).	Inflammation, surgical removal of lymph nodes (e.g., post-mastectomy).	**Lymphedema** (high protein)
Increased Capillary Permeability	Endothelial damage allows both fluid and protein to leak out.	Inflammation (acute), Trauma	**Exudate** (protein-rich,)

Hyperemia

Hyperemia is an active process resulting from increased blood inflow due to arteriolar dilation.

Cause: chemical mediators that cause vasodilation. It is often associated with inflammation or increased metabolic demand (e.g., exercising muscle).

Appearance: The tissue appears bright red (erythema) because the engorgement is due to oxygenated arterial blood.


Hyperemia
Normal blood flow
Arterioles
Hyperemia
Increased
blood flow
Cleveland
Clinic
©2023

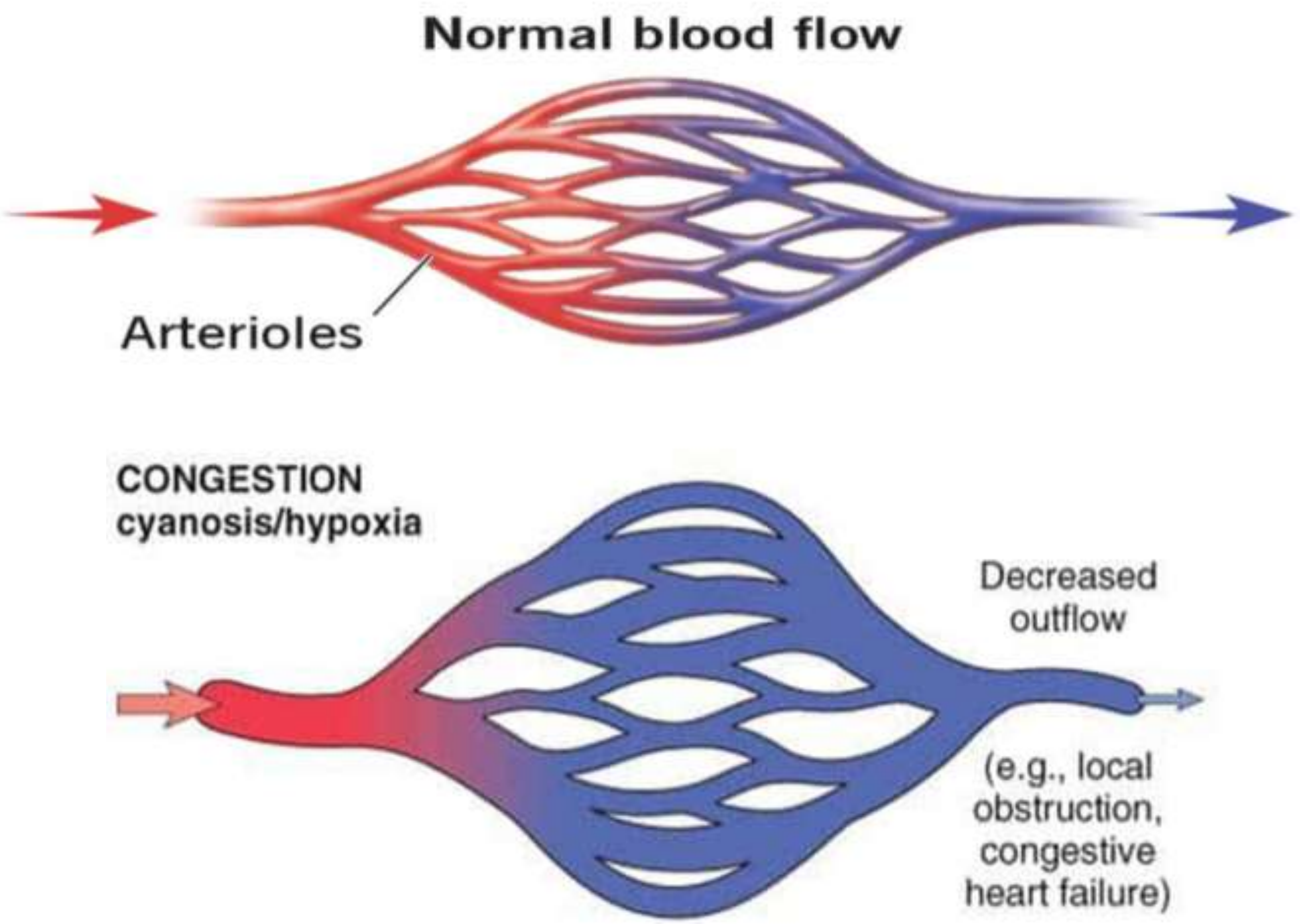
Congestion

Congestion is a passive process resulting from impaired venous outflow from a tissue.

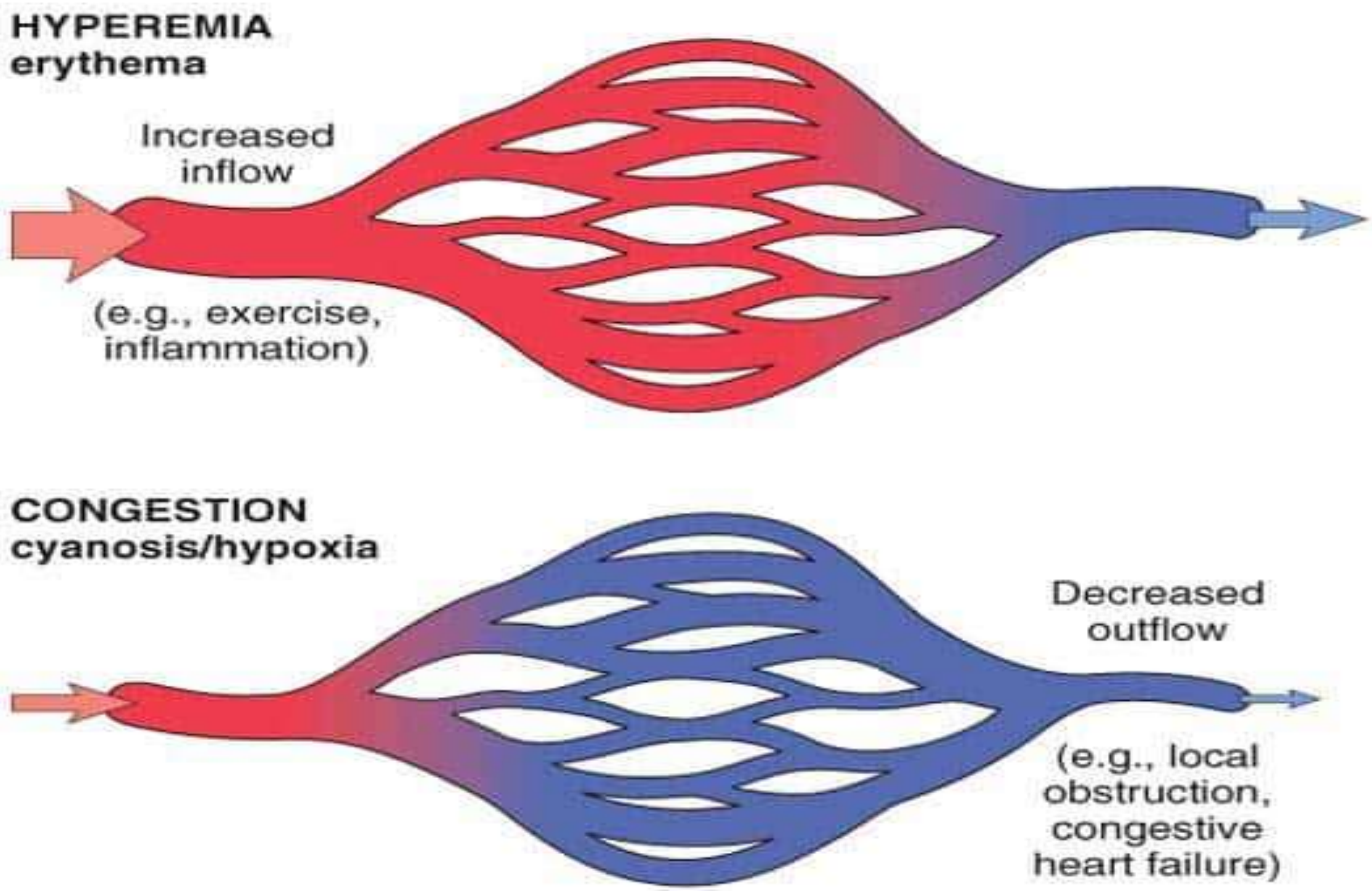
Cause: Systemic conditions like Heart Failure (causing widespread congestion) or local venous obstruction (like a Deep Vein Thrombosis).

Appearance: The tissue appears blue (cyanosis) because of the stasis of deoxygenated venous blood.

Hyperemia and Congestion



Hyperemia and Congestion



Hemorrhage

Hemorrhage is the extravasation (escape) of blood from the vessels into the surrounding tissues or spaces, or outside the body.

Causes:

- ❑ **Trauma**
- ❑ **Vessel Wall Defects**
- ❑ **Platelet or Coagulation Defects (Hemostatic Disorders):**

Hemorrhage

Hemorrhages are categorized by size and location:

- **Hematoma:** A large, localized collection of blood outside the blood vessels (e.g., a massive bruise).

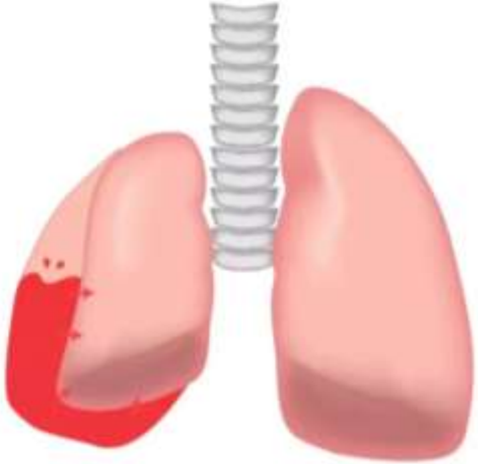
- **Petechiae (1-2 mm):** Tiny hemorrhages in the skin, mucous membranes, or serosal surfaces, usually platlet defect

Hemorrhage

- **Purpura (3-5 mm):** Slightly larger than petechiae, often or trauma.

- **Ecchymoses (1-2 cm):** Larger subcutaneous hemorrhages (bruises).

- **Hemothorax, Hemopericardium, Hemoperitoneum:** Large accumulations of blood in body cavities.




Thank you